

Simulating Relativistic Photography with Python

Patrick M. Mulikuza
Physics Department, Whitman College
Walla Walla, WA 99362
machumup@whitman.edu

Abstract

Einstein's theory of special relativity is one of the most famous yet counter-intuitive theories of modern physics. Part of this is because the predictions it makes are hard to observe in our everyday experience. In this experiment, we use Python to visualize a simulation of a book moving at relativistic speeds, providing an accessible illustration of key implications of Einstein's theory. We simulate a rectangular book moving on the x -axis, while a camera, triggered as soon as it aligns with the book, captures its image. At first, we simulate the book at rest using an 11×7 grid system. Then we visualize the book as seen from an observer who measures its velocity $\beta = 0.95$. Finally, we repeat the previous visualizations, accounting for light travel time to the camera. Comparing these three visualizations, we notice that the book's shape is twisted, not just squeezed. This is because the camera not only captures length contraction, but also captures light from different parts of the book emitted at different times, a phenomenon known as the Terrel-Penrose effect.

I. INTRODUCTION

In 1905, Albert Einstein published his theory of special relativity. One of the implications of this model is that close to the speed of light, an object's measured length is smaller than its proper length (length measured at rest relative to the object). This derives from the two postulates of Einstein's special relativity. First, the *principle of relativity*: the laws of physics are the same in all inertial (non-accelerating) reference frames. Second, the *constancy of the speed of light*: light travels at $c \approx 3 \times 10^8$ m/s in vacuum, regardless of the relative motion between source and observer.

These two postulates allowed Einstein to derive the Lorentz factor γ , which captures how relative motion stretches spacetime geometry:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (1)$$

where v is the relative velocity between source and observer, and c is the speed of light in vacuum.

Since $v < c$ for any massive object, $\gamma \geq 1$ always. As $v \rightarrow c$, $\gamma \rightarrow \infty$, and relativistic effects become extreme.

A. Derivation of Length Contraction

To get a better idea of how the length of objects moving near the speed of light contracts, consider two inertial reference frames (two non-accelerating reference frames): frame S (the lab frame, at rest) and frame S' (the moving frame, travelling at velocity v along the x -axis relative to S). Suppose a rod lies at rest along the x' -axis in S' . We want to find its length

as measured in S .

In S' (the rod's rest frame), the two endpoints of the rod are at positions x'_1 and x'_2 . We define an object's proper length as the length measured by an observer at rest with respect of the object. The rod's proper length is:

$$L_0 = x'_2 - x'_1 \quad (2)$$

Albert Einstein used Hendrik Antoon Lorentz's equations to convert spacetime coordinates between different inertial frames. The Lorentz transformation relates the space and time coordinates of an event measured in S to those measured in S' . For the x -coordinate:

$$x' = \gamma(x - vt) \quad (3)$$

To measure the length of the rod in S , we must record the positions of both endpoints at the same time in S , say, at time $t_1 = t_2 = t$. Applying equation (4) to each endpoint:

$$\begin{aligned} x'_1 &= \gamma(x_1 - vt) \\ x'_2 &= \gamma(x_2 - vt) \end{aligned}$$

Subtracting the first equation from the second:

$$x'_2 - x'_1 = \gamma(x_2 - x_1)$$

Substituting $L_0 = x'_2 - x'_1$ and $L = x_2 - x_1$:

$$L_0 = \gamma L \quad (4)$$

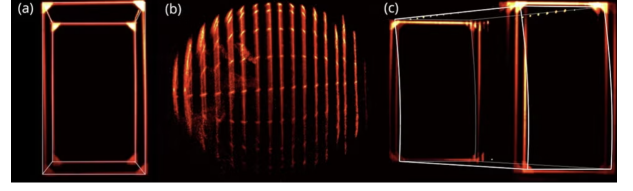
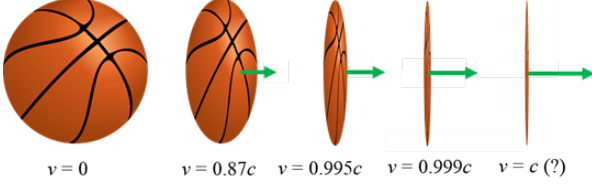


Figure 1. *Left:* Naive projection of the photograph of a ball moving at relativistic speeds, without accounting for the Terrell–Penrose effect. The ball appears squashed as its speed gets closer to c [6]. *Right:* First actual photograph of the Terrell–Penrose effect (2025), produced by researchers at TU Wien and the University of Vienna. A sphere at 99.9% the speed of light appears to have its north pole shifted, revealing its back side; a cube at 80% the speed of light appears rotated, showing multiple faces simultaneously [7].

Solving for L :

$$L = \frac{L_0}{\gamma} \quad (5)$$

Where:

- L is the contracted length.
- L_0 is the proper length (length in the rest frame).
- v is the relative velocity.
- c is the speed of light.
- $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$

This is the length contraction formula. Since $\gamma \geq 1$, it follows that $L \leq L_0$; the observed length is always less than or equal to the proper length.

B. The Terrell–Penrose Effect

For decades, physicists believed that an object travelling at relativistic speeds would appear squashed due to length contraction. Finally, in 1959, James Terrell and Roger Penrose corrected this misunderstanding by suggesting what we now know as the Terrell–Penrose effect [2, 3].

Accounting for the finite travel time of light from different parts of an object, they surmised the object would appear *rotated*, not compressed, because the camera captures photons emitted from different parts of the object at different times.

To illustrate this, consider a photon from the book travelling toward a stationary camera. Let x_{see} be the coordinate from which the photon must have been emitted to reach the camera at exactly $t = 0$. The contracted position of a rest-frame point x is x/γ , and since the book moves at $v = \beta c$, the photon was emitted at an earlier time $-t$ from:

$$x_{\text{see}} = \frac{x}{\gamma} - \beta t \quad (6)$$

The photon travels from (x_{see}, y) to the camera at $(0, 0)$, so $t^2 = x_{\text{see}}^2 + y^2$. Substituting Eq. (6) and solving for t (taking units where $c = 1$):

$$t = -\beta\gamma x + \gamma\sqrt{x^2 + y^2} \quad (7)$$

Substituting Eq. (7) back into Eq. (6) and using the identity

$$\frac{1}{\gamma} + \beta^2\gamma = \gamma:$$

$$x_{\text{see}} = \gamma x - \beta\gamma\sqrt{x^2 + y^2} \quad (8)$$

This shows that even a point at $x = 0$ appears displaced if the book has any vertical extent y , producing the visual illusion of rotation. Setting $\beta = 0$ recovers $x_{\text{see}} = x$, confirming the non-relativistic limit.

II. METHODS

We conducted all simulations using Python’s `numpy` [4] and `matplotlib` [5] libraries. The simulation proceeds in three stages: the book at rest, the Lorentz-contracted book without light travel time, and the apparent (photographed) shape accounting for finite light travel time. In all cases the book is modeled as an 11×7 grid of uniformly spaced points spanning $x \in [-5, 5]$ and $y \in [-3, 3]$ in the book’s rest frame. We set $\beta = 0.95$, giving a Lorentz factor $\gamma = 1/\sqrt{1 - \beta^2} = 1/\sqrt{1 - 0.95^2} \approx 3.20$.

A. Step 1: Book at Rest

Before applying any relativistic effects, we first establish what the book looks like at rest with no length contraction and no light travel time correction. We represent the book as a rectangular grid of 11×7 points, evenly spaced across $x \in [-5, 5]$ and $y \in [-3, 3]$, plotted inside a window spanning $x \in [-6, 6]$ and $y \in [-6, 6]$ so that all subsequent figures share the same coordinate system for direct comparison. The code generates this grid using `numpy.arange`, which creates evenly spaced values along each axis, and arranges them into matrices so that every combination of x and y corresponds to one grid point. `matplotlib` then plots each point as a small circle. Every distortion in the following steps is measured in comparison to this baseline shape.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 x_coords = np.arange(-5, 6, 1)
4 y_coords = np.arange(-3, 4, 1)
5 x_matrix = np.array([[x for x in x_coords] for _ in
6                       y_coords])
7 y_matrix = np.array([[y for _ in x_coords] for y in
8                       y_coords])
9 plt.figure()
10 plt.xlim(-6, 6)
11 plt.ylim(-6, 6)
```

```

10 plt.scatter(x_matrix, y_matrix, marker='o', s=20)
11 plt.xticks(np.arange(-6, 6.5, 0.5), rotation=45)
12 plt.yticks(np.arange(-6, 6.5, 0.5))
13 plt.xlabel("x_(book_frame)")
14 plt.ylabel("y_(book_frame)")
15 plt.title("Book_at_Rest")
16 plt.show()

```

Listing 1. Rest-frame grid

B. Step 2: Lorentz-Contracted Book

We then apply length contraction. As explained in the introduction, an observer watching a book move past them at velocity $v = \beta c$ will *measure* the book's length along the direction of motion to be shorter than its proper length by a factor of γ . In the code, this is implemented by dividing every x -coordinate by $\gamma \approx 3.20$, compressing the book's horizontal length. The y -coordinates are left unchanged, because Lorentz contraction only acts along the direction of motion. Note that this figure uses a smaller window of $x \in [-3.5, 3.5]$ and $y \in [-3.5, 3.5]$ rather than the $[-6, 6]$ window of the rest-frame plot, because the contracted grid no longer reaches beyond roughly $x \approx \pm 1.56$ and a tighter window makes the compressed shape easier to read. This is what a physicist would *measure* if they could record the positions of all points on the book simultaneously in the camera's frame. A real camera, however, records photons that arrive at the same time, not points that exist at the same time. Those photons left different parts of the book at different moments in the past, which is the correction we apply in Step 3.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 x_coords = np.arange(-5, 6, 1)
4 y_coords = np.arange(-3, 4, 1)
5 x_matrix = np.array([[x for x in x_coords] for _ in
6 y_coords])
7 y_matrix = np.array([[y for _ in x_coords] for y in
8 y_coords])
9 beta = 0.95
10 gamma = 1/np.sqrt(1-beta**2)
11 x_matrix = x_matrix * 1/gamma
12 plt.figure()
13 plt.xlim(-3.5, 3.5)
14 plt.ylim(-3.5, 3.5)
15 plt.scatter(x_matrix, y_matrix, marker='o', s=20, color
16 = 'green')
17 plt.xticks(np.arange(-3.5, 3.5, 0.5), rotation=45)
18 plt.yticks(np.arange(-3.5, 3.5, 0.5))
19 plt.xlabel("x_(camera_frame)")
20 plt.ylabel("y_(camera_frame)")
21 plt.title("Book_moving_along_the_x_axis")
22 plt.show()

```

Listing 2. Lorentz contraction along x

C. Step 3: Apparent Positions with Light Travel Time

We now compute where each photon *actually came from* to reach the camera at exactly $t = 0$ because a real camera captures a collection of photons that arrived at the same moment, each having left a different part of the book at a different point in the past. The apparent x -position of each point is given by Eq. (8), derived in the introduction:

$$x_{\text{see}} = \gamma x - \beta \gamma \sqrt{x^2 + y^2}$$

This shows that even points that share the same x -coordinate but sit at different heights on the book will appear at different horizontal positions in the photograph. The figure shares the same $[-6, 6]$ window as the rest-frame plot, so the leftward displacement of the apparent shape is directly comparable to the original book geometry.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 x_coords = np.arange(-5, 6, 1)
4 y_coords = np.arange(-3, 4, 1)
5 x_matrix = np.array([[x for x in x_coords] for _ in
6 y_coords])
7 y_matrix = np.array([[y for _ in x_coords] for y in
8 y_coords])
9 beta = 0.95
10 gamma = 1/np.sqrt(1-beta**2)
11 r0 = np.sqrt(x_matrix**2 + y_matrix**2)
12 x_apparent = gamma * x_matrix - beta * gamma * r0
13 plt.figure()
14 plt.xlim(-6, 6)
15 plt.ylim(-6, 6)
16 plt.scatter(x_apparent, y_matrix, marker='o', s=20,
17 color='red')
18 plt.xticks(np.arange(-6, 6.5, 0.5), rotation=45)
19 plt.yticks(np.arange(-6, 6.5, 0.5))
20 plt.xlabel("x_(camera_frame)")
21 plt.ylabel("y_(camera_frame)")
22 plt.title("Apparent_Shape_of_the_Book_(with_light_
23 travel)")
24 plt.show()

```

Listing 3. Apparent positions via Eq. (6)

III. RESULTS

A. Book at Rest

Fig. 2 confirms the expected undistorted geometry. The uniform Spacing and symmetric arrangement around the origin provide a clean reference against which relativistic effects are measured.

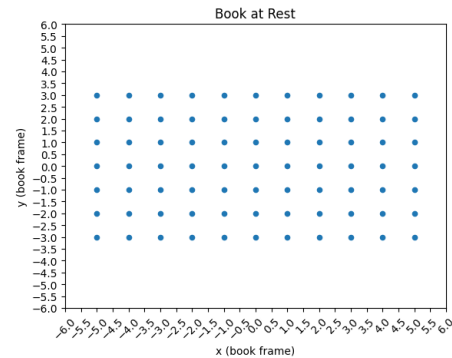


Figure 2. The 11×7 grid in the book's rest frame: a perfect rectangle spanning $x \in [-5, 5]$, $y \in [-3, 3]$.

B. Lorentz-Contracted Book

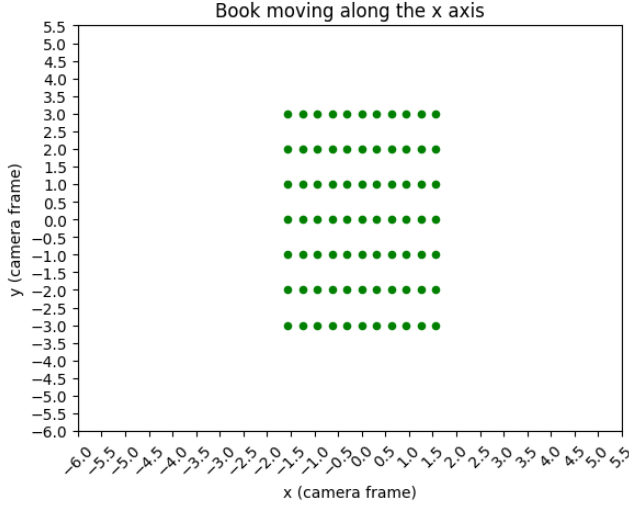


Figure 3. Lorentz-contracted book at $\beta = 0.95$: width compressed by $\gamma \approx 3.20$, from 10 to ≈ 3.1 units. Height unchanged.

Fig. 3 shows the grid after Lorentz contraction along x . The shape remains rectangular but is compressed to approximately 16.7% of its rest-frame width. This represents the *measured* length, not what a camera would photograph.

C. Apparent Positions with Light Travel Time

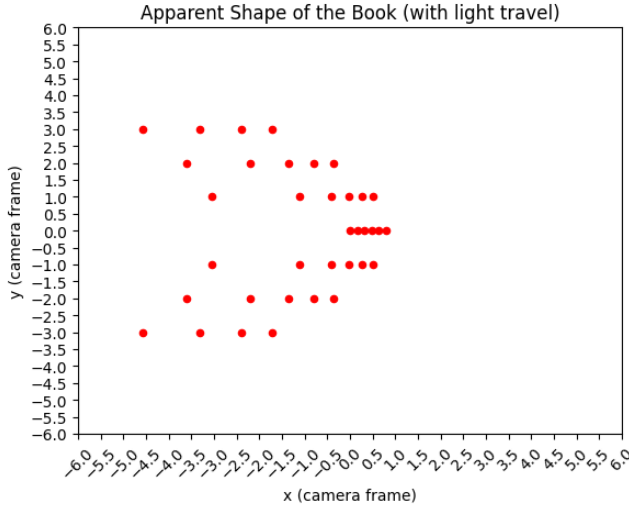


Figure 4. Apparent shape at $\beta = 0.95$ as recorded by the camera. The grid is sheared and stretched — the Terrell–Penrose rotation effect. Despite the white space, we maintain the same window size as the first simulation to allow comparison between the two and appreciate the rotation.

Fig. 4 reveals that the photographed book does not appear as a squashed rectangle. The shape appears rotated rather than

compressed, with the trailing edge shifted substantially further left than the leading edge.

IV. DISCUSSION

A. Rest Frame as Baseline

Fig. 2 confirms that our grid setup correctly represents a rectangular book. We use it as our baseline to evaluate the distortion in the subsequent figures.

B. Lorentz Contraction: Measurement vs. Appearance

At $\beta = 0.95$, the book is compressed to $\approx 16.7\%$ of its rest-frame width (Fig. 3). This shows what an observer moving relative to the book at $\beta = 0.95$ along the x –axis would see if all the photons that reach his camera departed from the book simultaneously. This, however, does not capture the reality of photography which involves photon travel time.

Looking at Fig. 2 and Fig. 3, we see that in the rest frame, the book spans from $x = -5$ to $x = +5$, giving a proper length of $L_0 = 10$ units. In the contracted plot, however, the leftmost dots sit at roughly $x \approx -1.56$ and the rightmost at $x \approx +1.56$, giving a contracted length of $L \approx 3.12$ units. The ratio is:

$$\frac{L}{L_0} = \frac{3.12}{10} = 0.312 \approx \frac{1}{\gamma} = \frac{1}{3.20}$$

Fig. 3’s shape perfectly aligns with the prediction of the length contraction equation.

Both Fig. 2 and Fig. 3 show the book spanning $y \in [-3, 3]$, a height of 6 units in both cases. The height is completely unchanged, confirming that length contraction only happens along the direction of motion.

The rest-frame book is a wide rectangle, roughly 10 units wide and 6 units tall, wider than it is tall. The contracted book, however, is a narrow rectangle, roughly 3.1 units wide and 6 units tall, now taller than it is wide. This shift perfectly illustrates length contraction. The internal spacing between dots has shrunk uniformly along x , meaning the contraction is perfectly uniform across the whole book, not just at the edges, which aligns with the length contraction formula, which multiplies the proper length by the same factor for every x -coordinate.

To estimate uncertainty, we read the dot positions from the plot visually, which introduces a reading error of ± 0.25 units on each side, following the convention that uncertainty is half the smallest division. Since the length involves two endpoints, the uncertainties add, giving ± 0.5 units on any length measurement.

Rest frame: $L_0 = 10.0 \pm 0.5$ units (from -5.0 ± 0.25 to $+5.0 \pm 0.25$). The ± 0.25 is a small uncertainty relative to the length itself, so this is a reliable reading. The true proper length lies somewhere between 9.5 and 10.5 units.

Contracted: $L = 3.12 \pm 0.5$ units (from -1.56 ± 0.25 to $+1.56 \pm 0.25$). The true contracted length lies between 2.62

and 3.62 units. This uncertainty is proportionally much larger than for the rest frame (± 0.5 on a length of 3.12 is a 16% uncertainty, compared to only 5% for the proper length), because the book is now so narrow that any error represents a bigger portion of the book's width.

The compression ratio is:

$$\frac{L}{L_0} = \frac{3.12 \pm 0.5}{10.0 \pm 0.5} = 0.312 \pm 0.051$$

This means the true ratio lies between 0.261 and 0.363. The predicted value $1/\gamma = 0.3125$ sits near the center of this interval, meaning our simulation is in excellent agreement with the theoretical prediction.

For the height, rest frame gives $H_0 = 6.0 \pm 0.5$ units and the contracted frame gives $H = 6.0 \pm 0.5$ units.

Uncertainty Propagation

The uncertainty on the ratio was propagated using:

$$\frac{\delta(L/L_0)}{L/L_0} = \sqrt{\left(\frac{\delta L}{L}\right)^2 + \left(\frac{\delta L_0}{L_0}\right)^2}$$

$$= \sqrt{\left(\frac{0.5}{3.12}\right)^2 + \left(\frac{0.5}{10.0}\right)^2} \approx 0.163$$

giving an uncertainty ratio of $0.312 \times 0.163 \approx 0.051$.

The bulk of this uncertainty comes from the contracted length measurement. The first term under the square root is about 64 times larger than the second, meaning that if we wanted a more precise ratio, the priority would be to find a better way to read off the contracted endpoints, not to improve the rest-frame measurement.

C. The Terrell–Penrose Effect

Comparing Fig. 4 and Fig. 2, the grid is no longer rectangular — it is sheared, with the dots on the right clustering tightly between $x \approx 0$ and $x \approx +1$. The book appears rotated rather than simply compressed.

In the apparent plot the rightmost dots sit at roughly $x \approx +1.0 \pm 0.25$ and the leftmost at roughly $x \approx -4.5 \pm 0.25$, giving an apparent total span of $L_{\text{app}} = 5.5 \pm 0.5$ units. The book still spans vertically from -3 ± 0.25 to $+3 \pm 0.25$, a distance of 6.0 ± 0.5 units. However, the total horizontal span alone is misleading because the shape is no longer rectangular — different rows of the grid have different widths. The whole shape appears shifted to the left because, in the camera's reference frame, the book is moving to the left, so all photons were emitted when the book was further to the left than it is at $t = 0$.

In the rest frame, every row, whether at $y = 0$ or $y = \pm 3$, spans exactly 10 units. In the photograph, however, the rows are no longer the same width. The center row at $y = 0$ spans only about 1 unit, from $x \approx 0$ to $x \approx +1.0$. The rows at

$y = \pm 3$, by contrast, span about 3 units, from $x \approx -4.5$ to $x \approx -1.5$. The top and bottom rows are therefore nearly three times wider than the center row.

The edges of the book appear stretched while the center appears compressed because the apparent width depends on $r_0 = \sqrt{x^2 + y^2}$, the distance from each dot to the camera. Dots at the top and bottom of the book ($y = \pm 3$) are farther from the camera than dots at the center ($y = 0$). Farther dots emitted their photons earlier, when the book was in a more leftward position, so the top and bottom rows get pulled further left, stretching their apparent width. The center row, being closest to the camera, emitted its photons most recently and is pulled left the least, making it appear narrow.

The book's shape is no longer a rectangle but a distorted figure where the top and bottom edges are displaced much further left than the center, due to differences in light travel time across the book. This produces the rotation-like distortion the camera captures. This is the Terrell–Penrose effect.

D. Limitations

Two simplifications bound the scope of this simulation. First, the book is modeled as a 2D grid of dots rather than a solid object, so the results show the correct geometry of the distortion but not photometric realism. Second, the simulation is planar, whereas the real Terrell–Penrose effect is fully three-dimensional; a 3D treatment would reveal additional rotational structure not captured here. Neither limitation affects the main conclusion.

V. CONCLUSION

While the Lorentz transformation predicts a physical length contraction along the direction of motion, this simulation highlights the crucial distinction between *measurement* and *visual appearance*. Due to the finite speed of light, photons from the trailing edge of the moving book arrive later than photons from the leading edge, causing the observer to see the side of the book even when it faces the camera directly. As a result, the object does not appear flattened; it appears rotated in three-dimensional space. The Terrell–Penrose effect proves that what an observer sees with a camera is a distorted projection of relativistic contraction, reconciling the geometry of special relativity with the realities of light-signal transmission.

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